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Presenting your AI in education research with rigor: The TEP-AIED model

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ABSTRACT

The rapid expansion of Artificial Intelligence in Education (AIED), particularly with the emergence of generative AI, has intensified the need for rigorous, transparent, and pedagogically grounded research reporting. Although existing frameworks offer valuable guidance on ethical responsibility or technical disclosure, they may be complex, fragmented, or difficult to operationalize in empirical and experimental contexts. This paper proposes the TEP (Transparency-Ethics-Pedagogy)-AIED model, a streamlined three-dimensional framework integrating transparency, ethics, and pedagogy to guide the design and reporting of AIED studies. The model highlights clear disclosure of AI system characteristics and learner/teacher-AI interaction processes to strengthen interpretability and reproducibility, and it encourages proactive ethical governance by addressing data handling, risk mitigation, equity, and learner agency. It also foregrounds pedagogical grounding by requiring explicit articulation of learning objectives, theoretical alignment, and learner preparation to develop appropriate conceptions of AI-supported learning and avoid uncritical reliance. By positioning these dimensions as interconnected rather than as isolated requirements, the TEP-AIED model supports methodological rigor while remaining accessible to researchers and practitioners. Intended as practical guidance for reporting AIED research, the framework encourages authors to embed transparency, ethics, and pedagogy throughout the research lifecycle, thereby strengthening validity, accountability, and educational relevance in AI-enhanced learning research.

1. Trends of artificial intelligence in education

Owing to the advancement and popularity of artificial intelligence technologies, the amount of Artificial Intelligence (AI) in Education (AIED) research has rapidly increased in recent years (Allison et al., 2025). In particular, the emergence of Generative Artificial Intelligence (GenAI) has attracted widespread discussion among researchers in education and educational technology (Tu & Lu, 2025). Many studies have reported the effectiveness of applying AI to educational settings and professional training in various domains, such as language education (Chang et al., 2025; Chen et al., 2025; Zhang et al., 2025), science and mathematics education (Yi et al., 2025), physical education (Hsia et al., 2025), art education (Chiu & Hwang, 2025), social science education (Hwang et al., 2025), programming education (Wang et al., 2025), and nursing education (Chang & Hwang, 2025; Chang & Su, 2025).

Scholars have also discussed the impacts of using AI (and particularly GenAI) in educational settings, including its influence on learners' metacognitive laziness and cognitive offloading (Fan et al., 2025). Cognitive offloading refers to the strategic transfer of mental effort from individual learners' internal cognitive resources to external tools or supports to reduce working memory demands and manage task complexity (Chirayath et al., 2025; Richmond & Taylor, 2025; Risko & Gilbert, 2016). Metacognitive laziness, in contrast, describes a diminished engagement in metacognitive regulation, such as monitoring, planning, and evaluating the accuracy of information, as well as adjusting strategies, particularly when external tools are available (Fan et al., 2025). From an AIED perspective, cognitive offloading can function as a productive learning strategy when accompanied by active metacognitive control, enabling learners to allocate cognitive resources toward deeper reasoning; however, when metacognitive monitoring and regulation decline, offloading may evolve into metacognitive laziness, potentially undermining conceptual understanding, transfer, and learner

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autonomy.

Therefore, several scholars have indicated the need to conduct AIED research to examine not only learners' performance outcomes, but also how they are guided to engage with AI tools (Fan et al., 2025; Tu, 2024; Weidlich et al., 2025). That is, when conducting and reporting an AIED study, it is important for researchers to present the details of how the learners interact with AI during the learning process, as well as the aims and procedures of designing and implementing the research. This requirement implies that reporting an AIED study needs to take several dimensions into account, including transparency, ethics, and pedagogy, in order to ensure the external validity and pedagogical accuracy of the research.

2. The TEP-AIED model: transparency, ethics, and pedagogy for AIED research

Several scholars have proposed guidelines or frameworks to help researchers conduct and present their research designs and findings of AIED studies. For example, Fu and Weng (2024) proposed a human-centered responsible AI framework for education that foregrounds ethical safeguards, learner agency, and data governance in AI-supported learning environments; Topali et al. (2025) proposed a pedagogical considerations framework for AIED in authentic K-12 settings for examining how AI technologies are integrated into real classroom environments rather than controlled laboratory contexts. This framework emphasizes three interrelated dimensions: theoretical alignment (i.e., AI tools must be explicitly grounded in established learning theories, such as scaffolding, self-regulated learning, and formative feedback, rather than being adopted for their technological novelty) and contextual sensitivity (i.e., classroom realities, such as curriculum constraints, teacher expertise, student diversity, and institutional infrastructure, shape how AI systems function in practice). The RAISE (Reporting AI Studies in Education) framework proposed by Allison (2026) provides structured and transparent reporting guidelines for AIED research. It provides a comprehensive checklist designed to guide authors in clearly documenting the design, implementation, and evaluation of AI-supported interventions.

The human-centered responsible AI framework of Fu and Weng (2024) provides a strong ethical foundation, which emphasizes fairness, accountability, privacy, and stakeholder responsibility; however, it remains primarily ethics-centric and less explicit about how such principles translate into concrete research design and reporting practices in AIED. In contrast, Topali et al. (2025) foreground pedagogical grounding and contextualization in authentic K-12 settings, addressing alignment with learning theories and classroom realities; however, this framework is largely pedagogically oriented and does not systematically integrate transparency or procedural reporting requirements. The RAISE framework (Allison, 2026) offers a highly comprehensive reporting checklist covering technical, pedagogical, and ethical aspects, but its breadth and granularity may make it complex and less accessible for routine empirical applications. That is, although these existing frameworks are helpful to researchers in preparing and reporting AIED research, certain limitations may hinder their practical adoption. These observations suggest the need for a new reporting model that maintains theoretical and ethical rigor while offering clear, streamlined structures that support accessibility, coherence, and practical applicability in AIED research.

Recent work has also highlighted several methodological and conceptual problems that make many GenAI studies difficult to interpret. Weidlich et al. (2025) argued that claims about the effects of ChatGPT on learning are often undermined by insufficiently specified treatments, poorly matched or weakly described control conditions, and outcome measures that do not validly represent learning. They further noted that ChatGPT is a tool rather than a pedagogy, meaning that its educational value depends on how it is embedded within an instructional design rather than on the tool itself. Yan et al. (2025) raised a closely related concern. They argued that research on GenAI often fails to distinguish between immediate task performance and learning as a durable change in knowledge and capability, and called for greater attention to retention, transfer, delayed recall, and the cognitive and metacognitive processes that underlie learning. These concerns point to a basic issue in current AIED research: studies are often not sufficiently clear about what kind of outcome is being measured and what that outcome actually tells us about learning. A reporting framework should therefore require clearer specification of instructional design, comparison conditions, learner-AI interaction, and the rationale for the learning measures that are used.

In this paper, we propose a three-dimensional TEP model (i.e., transparency, ethics, and pedagogy) for guiding researchers to prepare and present AIED research, as shown in Fig. 1. The TEP-AIED (TEP in AIED research) model is grounded in recurring limitations identified in existing AIED research and frameworks. Transparency is essential because AI-supported interventions are often insufficiently specified, making it difficult to interpret findings or replicate studies; clear reporting of system characteristics, prompts, and learner-AI interactions addresses this gap (Allison, 2026; Weidlich et al., 2025). Ethics is emphasized due to the growing concerns surrounding data governance, bias, equity, and learner agency, all of which are central to responsible AI integration in educational contexts (Fu & Weng, 2024). Pedagogy is equally critical because AI functions as a tool rather than an instructional approach, and its educational value depends on alignment with learning objectives, theoretical foundations, and meaningful learner engagement (Topali et al., 2025; Weidlich et al., 2025). By foregrounding these three dimensions together, the model responds to the need for a framework that is not only conceptually rigorous but also practically applicable, integrating methodological clarity, ethical responsibility, and pedagogical validity within a single coherent structure.

In the TEP-AIED model, transparency involves clear reporting of system specifications, prompt design, and reproducibility considerations to ensure interpretability and methodological clarity. Ethics encompasses data governance, risk mitigation, and equity and accessibility safeguards to support responsible AI integration. Pedagogy foregrounds learning objectives, theoretical alignment, the instructional role of AI, and learner orientation to ensure meaningful and developmentally appropriate engagement with AI tools. It also requires explicit consideration of the teacher's role in AI-supported environments, particularly in K-12 contexts, where teachers are expected to model appropriate AI use (e.g., demonstrating how to prompt or evaluate outputs), monitor student-AI interactions, and scaffold learners' critical engagement with AI-generated content. In this sense, AI is positioned as a tool embedded within teacher-guided instructional processes rather than as a replacement for pedagogical decision-making.

In practice, the three dimensions, including transparency, ethics, and pedagogy, are mutually reinforcing rather than independent. For example, disclosing prompts, system settings, and learner-AI interaction processes under Transparency not only improves interpretability and reproducibility, but also helps researchers identify ethical risks such as bias, hallucination, or overreliance, while enabling pedagogical reflection on whether the AI support aligns with the intended learning objectives and level of learner engagement. It is expected that researchers can explain how AI meaningfully contributes to learning processes and outcomes in helping learners think critically from diverse angles, rather than completing the learning tasks for learners.

In comparison with the existing frameworks or models, the TEP-AIED model is a more streamlined and integrative alternative; it synthesizes transparency, ethics, and pedagogy into three interdependent dimensions, reducing complexity while maintaining conceptual rigor and providing clearer guidance for both study design and reporting. This positions TEP-AIED as a more operational and accessible framework for advancing methodological clarity and educational validity in AIED research.

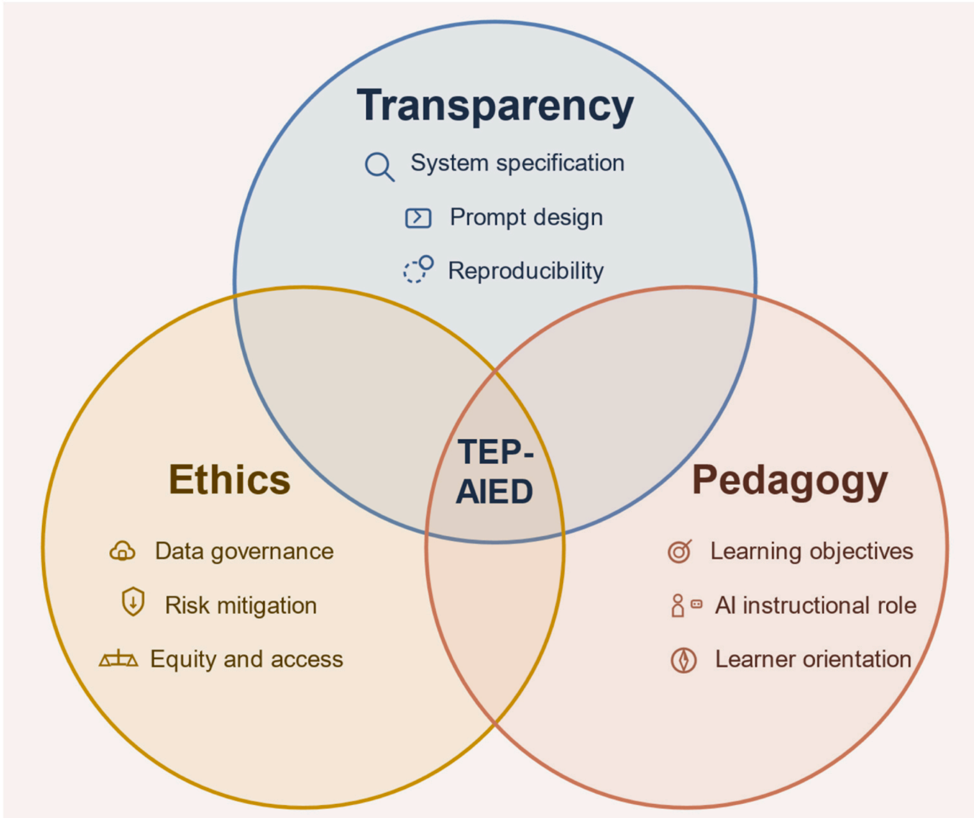

Fig. 1. The TEP (Transparency-Ethics-Pedagogy)-AIED model.

3. Guidelines based on the TEP-AIED model

When reporting the research design, implementation, and findings, it is suggested that researchers refer to the TEP-AIED guidelines in Table 1. The guidelines should be consulted during the research design stage, rather than being applied only at the manuscript preparation stage.

In comparison with existing frameworks, the TEP-AIED model offers a more streamlined and design-oriented structure that balances rigor with usability. While comprehensive checklists provide detailed reporting guidance and ethics-focused frameworks foreground responsible AI practices, they may be perceived as complex or difficult to operationalize in empirical research contexts. The TEP-AIED model complements these efforts by synthesizing key dimensions into a clear three-part structure that is easier to apply during both the research design and reporting stages. By explicitly integrating pedagogical calibration, technical transparency, and ethical accountability within a single coherent framework, the model provides practical guidance without overwhelming researchers, thereby enhancing its accessibility and potential for broader adoption in AIED studies.

To clearly address that a study has been conducted and reported following the model, it is suggested that the authors add a subsection to the Method or Research Design section titled "Transparency, Ethics, and Pedagogy Considerations of Using AI" to address how the guidelines were implemented by referring to the TEP-AIED model. An example is provided below.

"Transparency, Ethics, and Pedagogy Considerations of Using AI

*The study was designed and reported following the guidelines of the TEP-AIED model **(Citation)**. The TEP-AIED model pays explicit attention to transparency, ethics, and pedagogy in the use of AI. In terms of transparency, the study clearly documents the AI tool employed, its version and provider, the instructional workflow through which students engaged with it, the prompts or guidance used, and, where applicable, the differences between treatment and comparison conditions while keeping other instructional elements constant. From an ethical perspective, the study reports ethical approval and informed consent procedures, explains how participant data were anonymized, stored, and managed, and addresses potential risks associated with AI use, such as hallucination, bias, and student overreliance, together with relevant accessibility and equity considerations. Pedagogically, the study specifies the intended learning objectives, research hypotheses, and the learning theories or instructional principles underpinning the intervention, while also clarifying the instructional role of AI in the learning process. In addition, learners were oriented before the intervention regarding appropriate AI use, acceptable and unacceptable practices, and strategies for verifying and reflecting on AI-generated outputs. These considerations guided both the design of the intervention and the interpretation of outcomes related to performance, retention, or transfer, thereby ensuring that the educational and technological contributions of the study were aligned with the TEP-AIED framework."*

In addition, although not mandatory, authors are strongly encouraged to provide links to the source code, tools, datasets, or benchmarks used in their studies whenever possible. Such practices can enhance transparency, reproducibility, and opportunities for further validation and extension of AIED research. In cases where materials cannot be shared openly, authors should clearly explain the reasons, such as privacy, licensing, or institutional restrictions.

4. Conclusions

The TEP-AIED model is a streamlined, three-dimensional framework for guiding the design and reporting of AIED research. By integrating transparency, ethics, and pedagogy into a coherent structure, the framework contributes to the field in three key ways. First, it emphasizes technical and procedural transparency to enhance the interpretability and reproducibility of AI-supported interventions. Second, it

Table 1
TEP-AIED guidelines.

Dimension	Paper section	Guidelines
Transparency	Learning approach	1 Provide interface screenshots (or a schematic) of the AI tool. 2 Describe student workflow steps (when/where AI is used). 3 Specify prompts or guidance given to learners (in an appendix if the illustration is long). 4 If a/an (quasi-)experimental study, clearly define the treatment condition and the comparison/ control condition, including how AI use differs across conditions and what instructional activities are held constant.
	Adopted system or system development	1 Adoption of existing AI system: Model/system name, version, and provider (e.g., ChatGPT3.5 by OpenAI) 2 Development of new AI system: system architecture, design rationale, training data, training process, algorithmic configuration, test data, and validation process.
Ethics	Participants	Report ethical approval and consent procedures.
	AI Governance and Risk Considerations (where applicable)	1 Report data governance (data sharing, storage, retention, anonymization). 2 Describe risk mitigation strategies (e.g., bias, hallucination, harmful outputs, and overreliance). 3 Explain equity and accessibility measures (language, disability support, and access gaps). If not applicable, provide justification. 4 Explain why the data/benchmarks used are significant and broad. If they are not open-source, clearly explain the data used and the reasons why they might not be available (in accordance with the data sharing policy of most well-recognized journals).
Pedagogy	Introduction (background and motivation)	1 Specify learning objectives. 2 State the research hypotheses and their justifications. 3 Address the learning challenge the study addresses and identify the learning theories or instructional principles guiding the AI-supported intervention. 4 Explain the instructional role assigned to the AI (e.g., tutor, feedback agent, co-writer) and the targeted form of learning.
	Experimental process	1 Report how learners were oriented to appropriate AI use before the intervention, including the intended role of AI, acceptable vs. unacceptable use, and verification/reflective strategies. 2 Describe the detailed procedures through which participants used AI to achieve the intended learning goals, including the sequence of activities, prompts/ interactions, and how AI use was embedded in the instructional process. 3 Generalization to data unseen in learning and outside the domain studied is important to demonstrate the validity of learning. 4 Specify whether outcomes assess performance, retention, or transfer.
	Discussion	1 Justify outcome measures aligned with those objectives (e.g., conceptual understanding, transfer, self-regulated learning). 2 Explain the significant innovations made, whether in AI or educational aspects, and whether the results are generalizable to a broader domain.

foregrounds ethical responsibility beyond formal approval, encouraging researchers to reflect on data governance, risk mitigation, equity, and learner agency. Third, and most distinctively, it positions pedagogy at the center of AIED research by requiring explicit articulation of learning objectives, theoretical grounding, learner preparation for appropriate AI use, and generalization of learned results. In doing so, the model bridges the gap between technical reporting standards and pedagogically meaningful experimentation, offering a balanced alternative to overly complex checklist-based approaches.

The TEP-AIED model is particularly applicable to experimental and intervention-based AIED studies, where clarity in system configuration, learner-AI interaction, and instructional design directly affects internal and external validity. However, its principles may also inform qualitative, design-based, and system-development research by encouraging reflective alignment between AI functionality and educational purpose. Researchers are encouraged to consult the framework at the research design stage rather than only during manuscript preparation, ensuring that considerations of transparency, ethics, and pedagogy are embedded throughout the study lifecycle. By adopting this integrated perspective, AIED scholars can contribute to a more rigorous, responsible, and pedagogically grounded research culture in the evolving landscape of AI-enhanced learning.

CRediT authorship contribution statement

Gwo-Jen Hwang: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Conceptualization. **Haoran Xie:** Writing – review & editing, Conceptualization. **Benjamin W. Wah:** Writing – review & editing, Conceptualization. **Dragan Gašević:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors, Gwo-Jen Hwang, Haoran Xie, Benjamin W. Wah, and Dragan Gašević, are Editors-in-Chief for this journal and were not involved in the editorial review or the decision to publish this article.

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